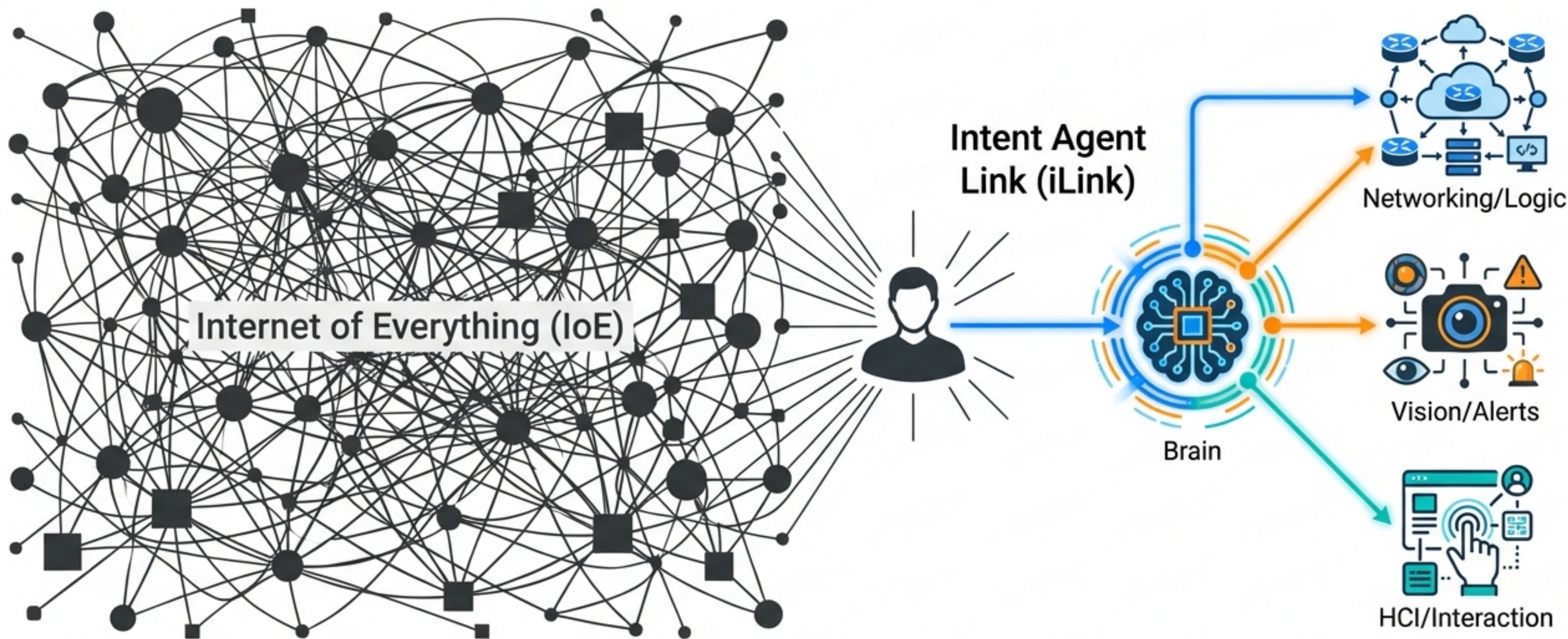


From Internet of Everything (IoE) to Intent Agent Link (iLink)

Pioneering Intent-Driven Intelligence in Networking, Computer Vision, and HCI.



The Vision: Intent-Driven Closed Loop

Moving beyond connecting devices (IoE) to connecting intelligent agents. A theoretical framework based on Calculation—Deployment—Learning—Understanding—Interaction.

Team Profile

Lead: Prof. Jingyu Wang (CIC Fellow).
Dual Background: Academic Rigor + Industrial Scale.



300+ Top-Tier Papers
(NSDI, SIGCOMM, CVPR)



**National Science and
Technology Progress
Award (2nd Prize)**



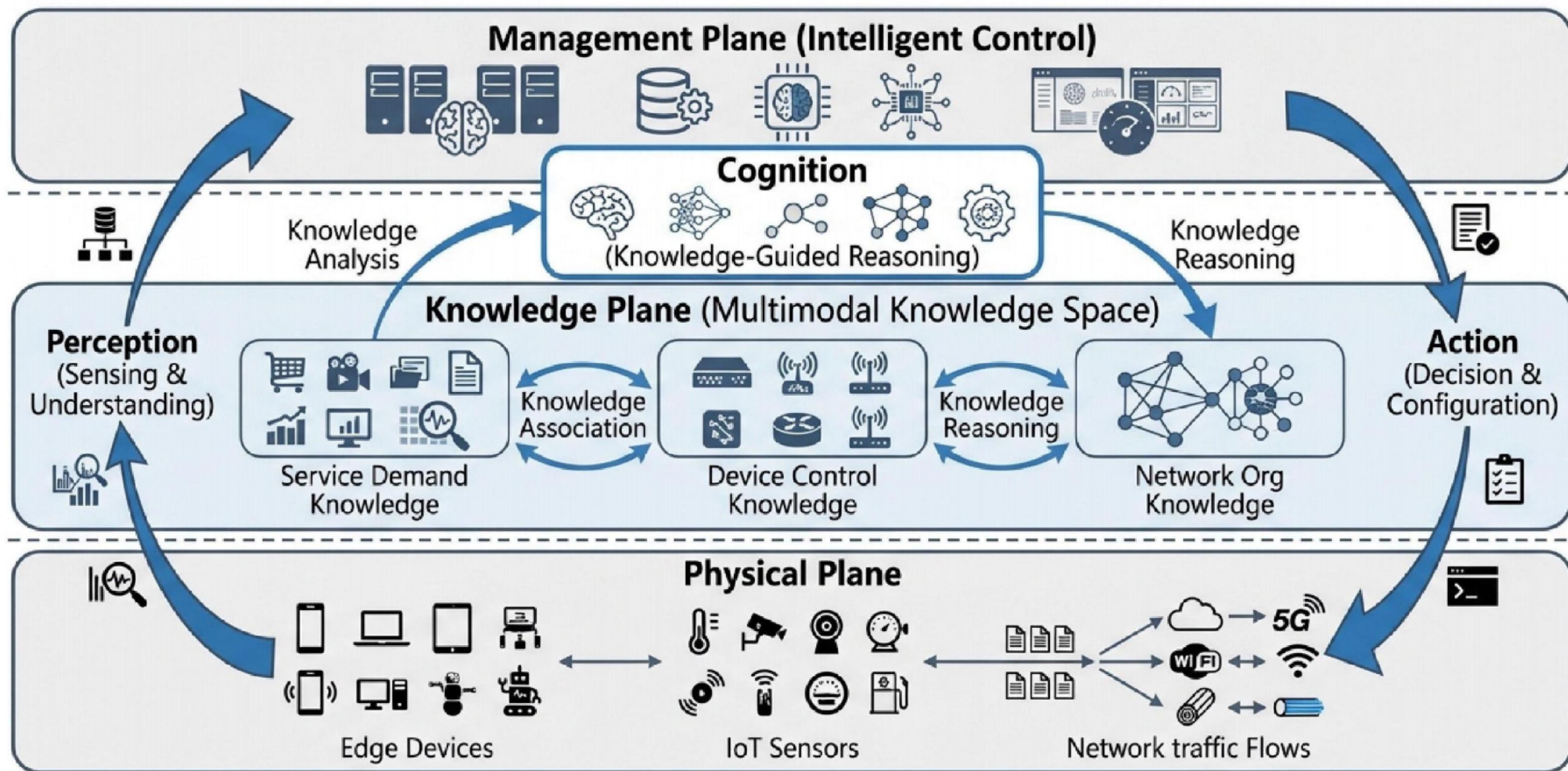
**Deployed in 15+ Provincial
Units**



Open Source Contributions:
OpenNetAI

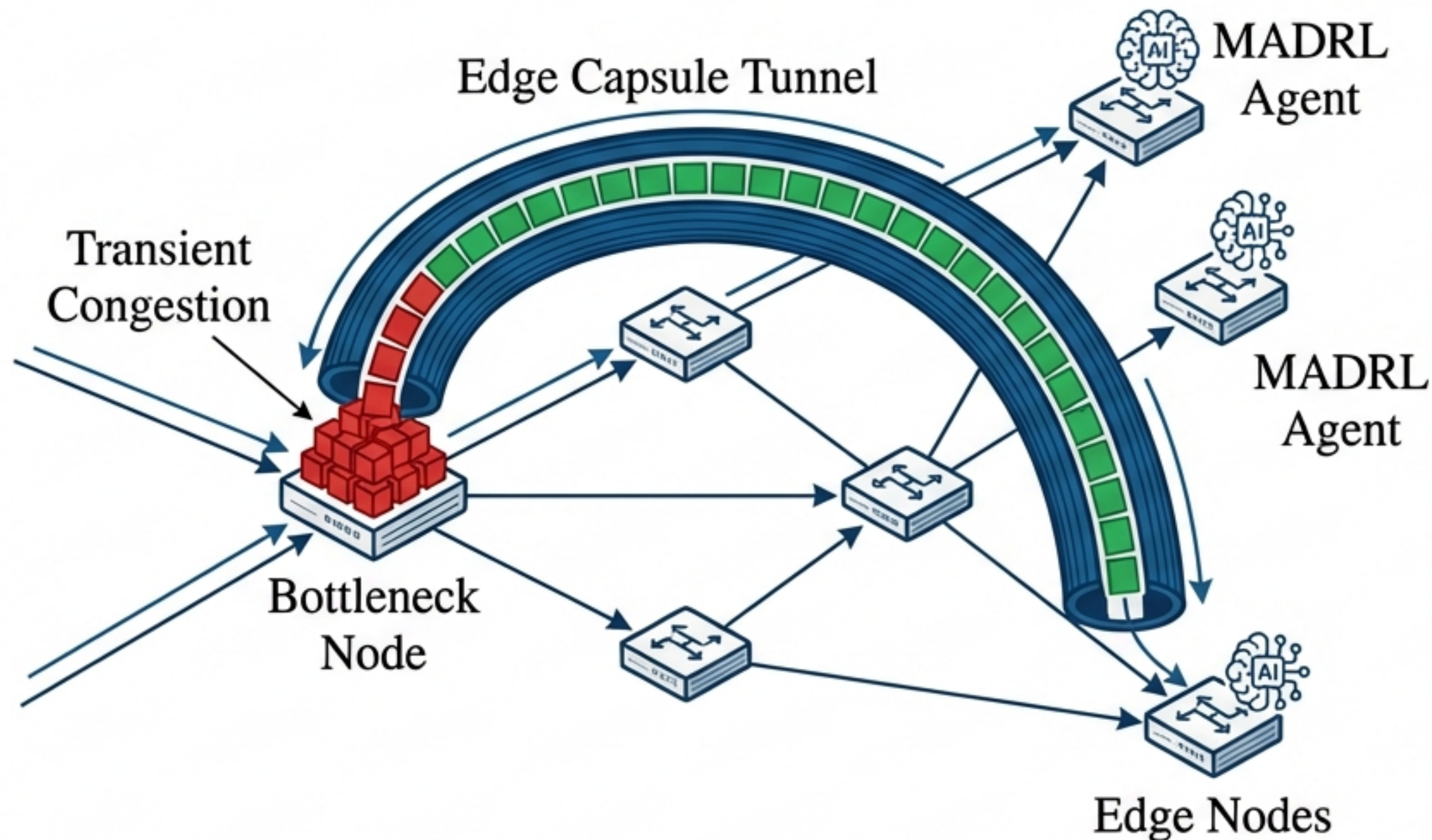
Knowledge-Defined Intent-Based Networking (AI4Net)

Innovation Point:
6G-ADM Architecture
& Network Copilot



HyperWay: The Nervous System of Massive IoT

Proactively Mitigating Transient Congestion via Edge Capsule Tunnels

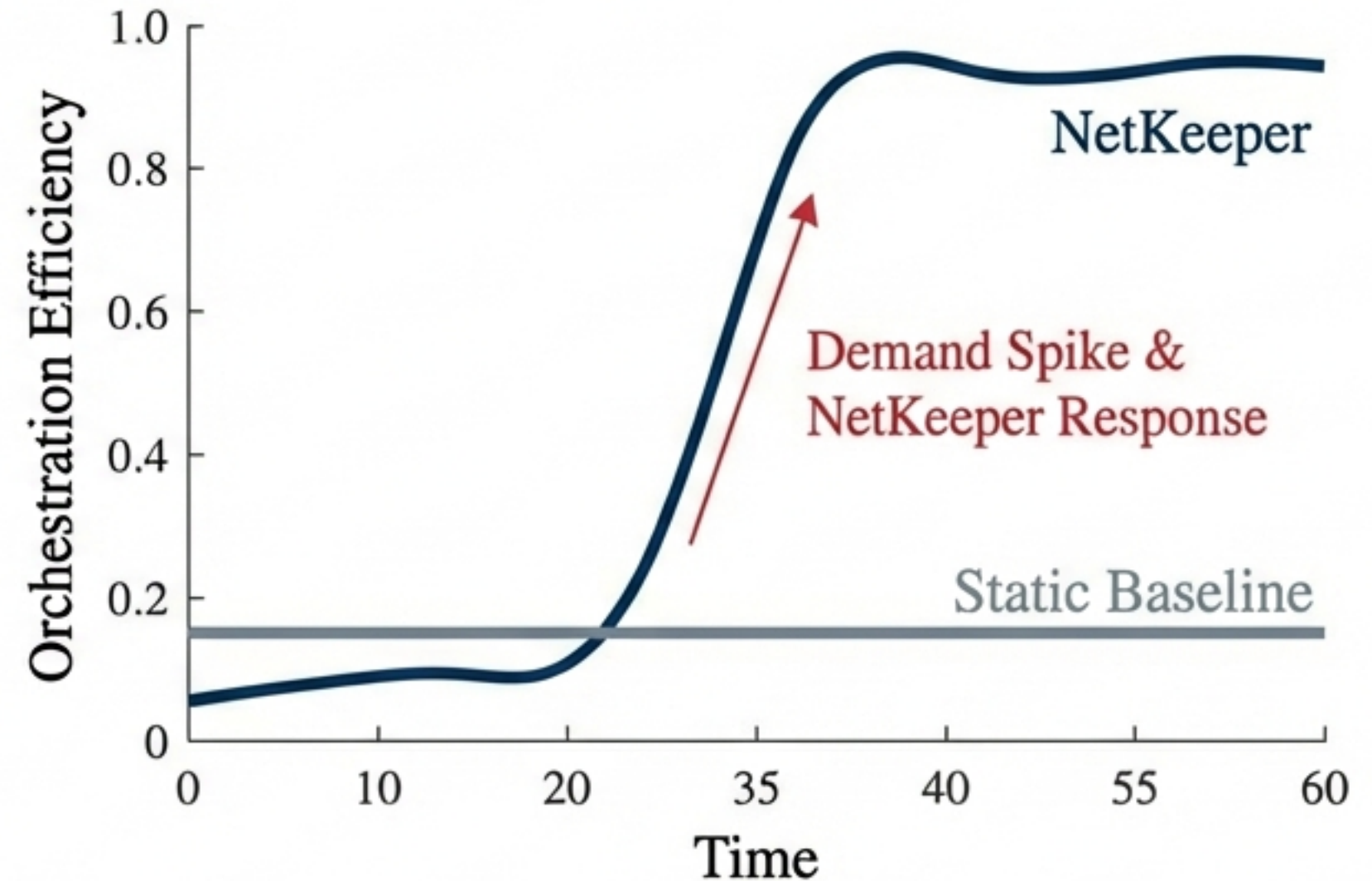


- **Challenge:** Packet loss in time-critical IoT.
- **Solution:** Edge Capsule Tunneling + MADRL.
- **Mechanism:** Agents predict congestion and reroute via encapsulated tunnels.
- **Source:** ACM MobiCom '24.

Autonomous Orchestration & Network Resilience

Innovation:
NetKeeper & Cold-Start Orchestration

- **NetKeeper:** Autonomous configuration updates based on traffic patterns.
- **Cold-Start Slice:** Cross-Slice Federated Meta Learning.
- **Congestion Control:** QUIC-enabled framework for time-critical IoT.

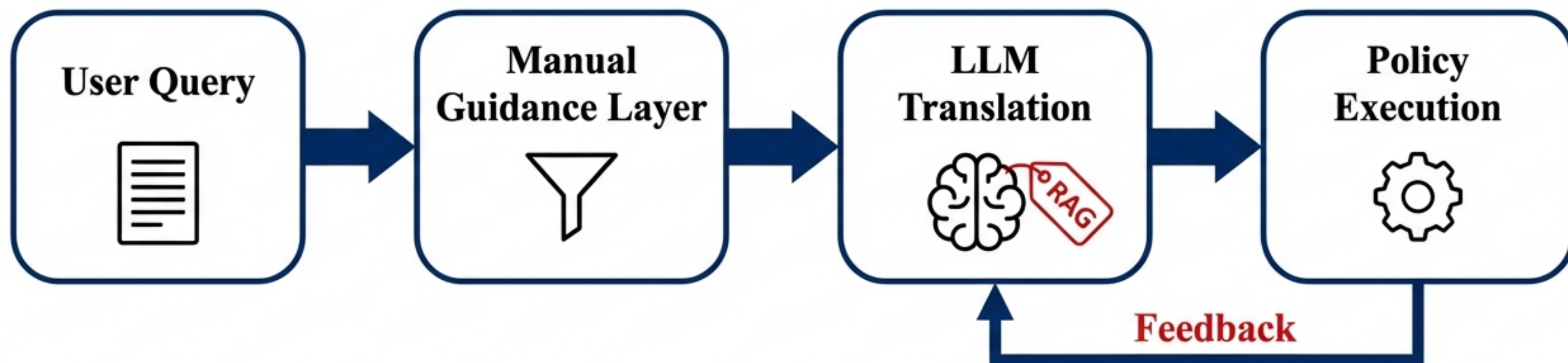


Schematic graph showing NetKeeper's adaptive response compared to static baseline.

LLM-Empowered Intent Translation & Policy Solving

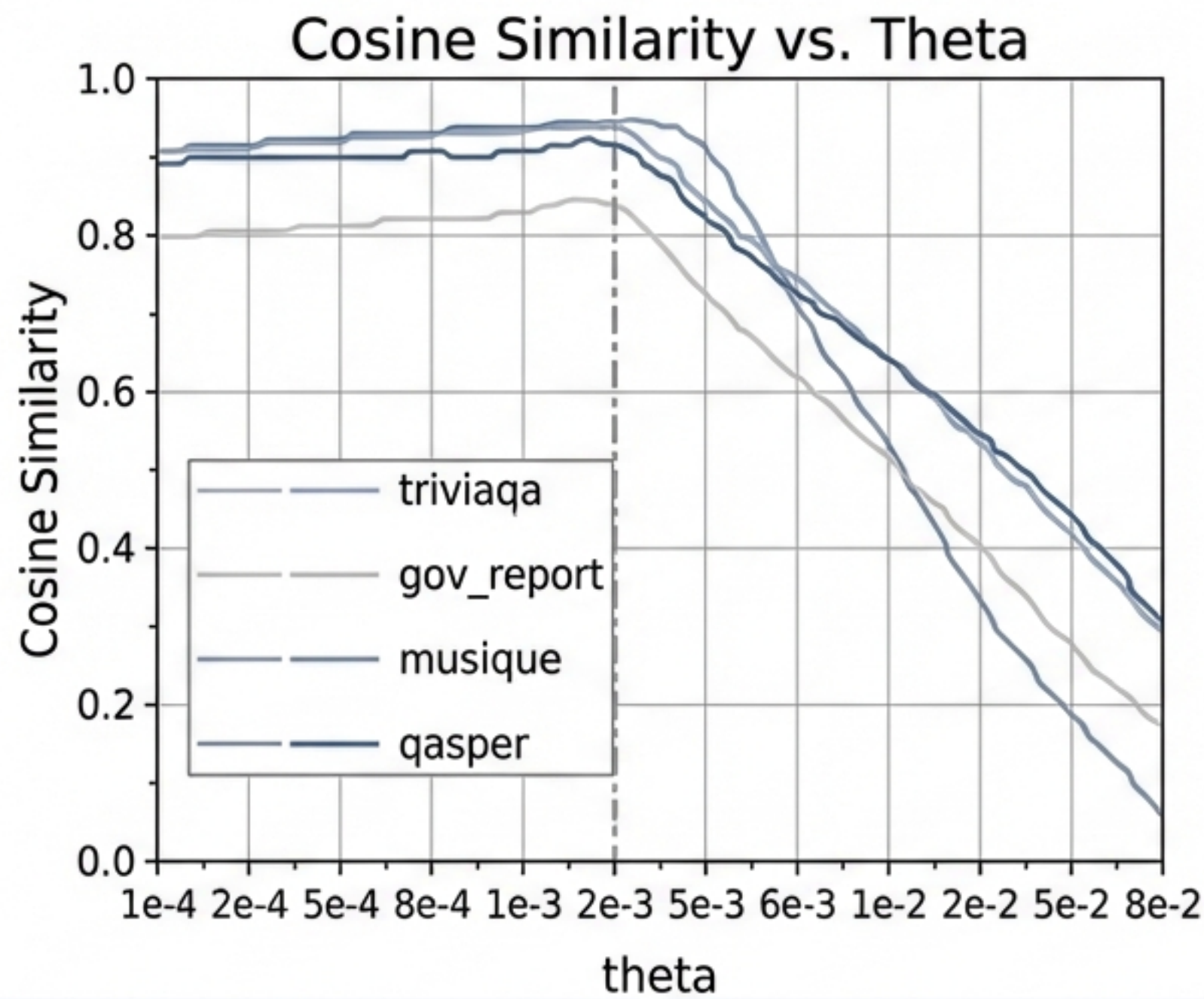
Innovation Point: Compass
(RAG-based Translation)

The Compass Workflow

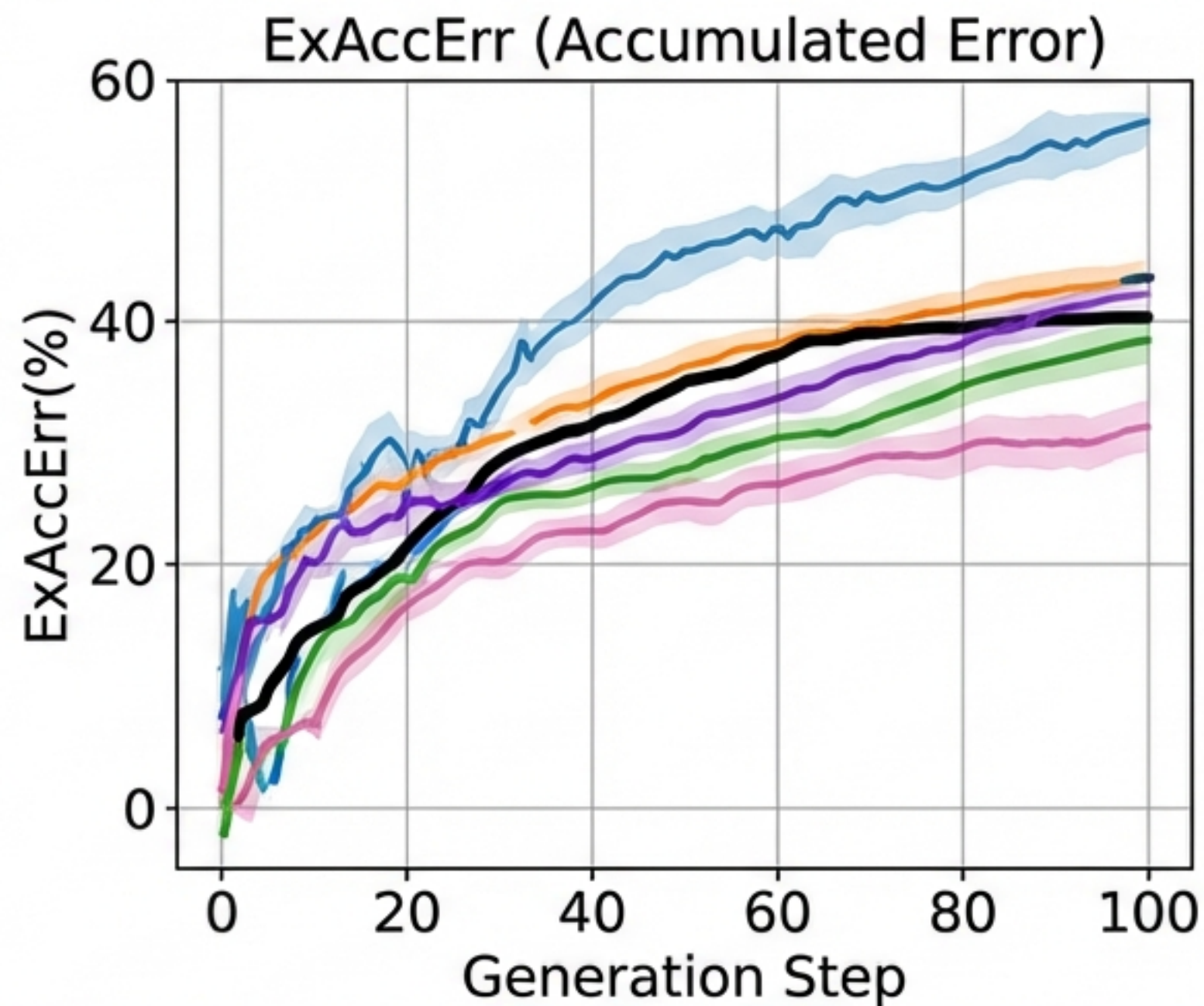


- Mitigating hallucination via hierarchical index retrieval.
- Fast Access Control List (ACL) policy solving using Graph Neural Networks.

System Efficiency: Foundation Models on the Edge



Innovation: Gradient Sparsification & ExAccErr Metric



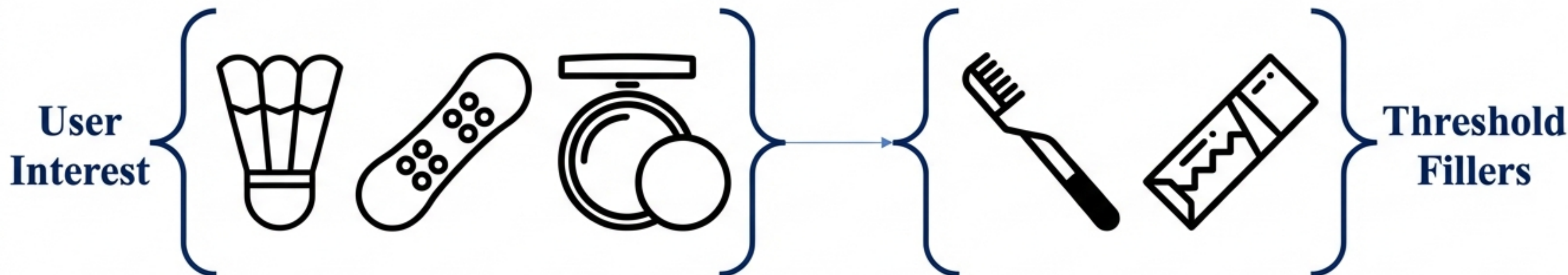
mTopk memory mechanism compensates for sparsification errors, allowing massive models to run on constrained edge devices.

Multimodal Learning: Cross-Modal Alignment

Innovation:

Deep Graph Alignment &
Threshold-Meeting Behavior

Shopping Basket Analysis

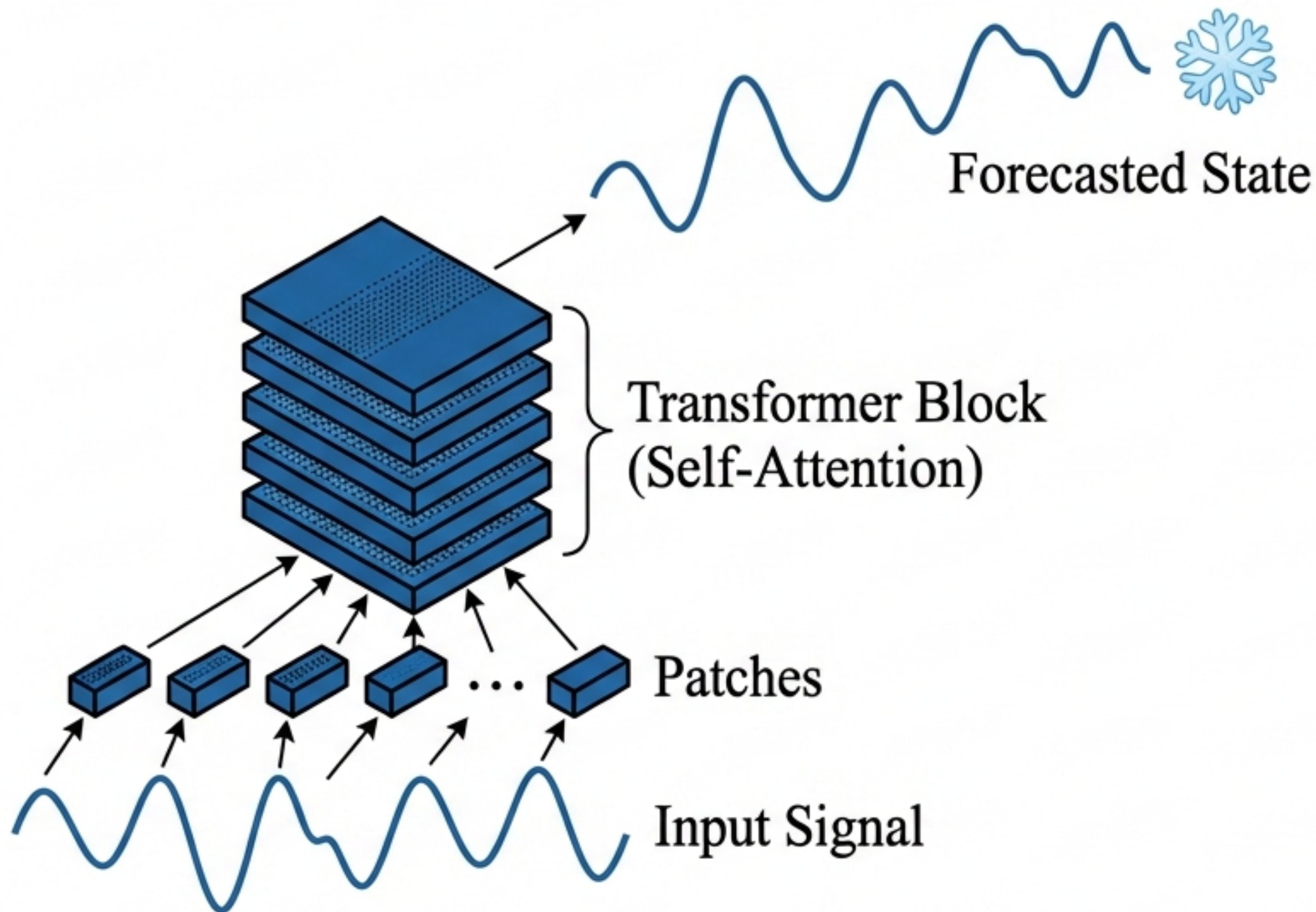


Aligning disparate data sources to distinguish **genuine user interest** from **‘threshold-meeting’** economic behavior (e.g., adding items for free shipping).

Time-Series Representation & Forecasting.

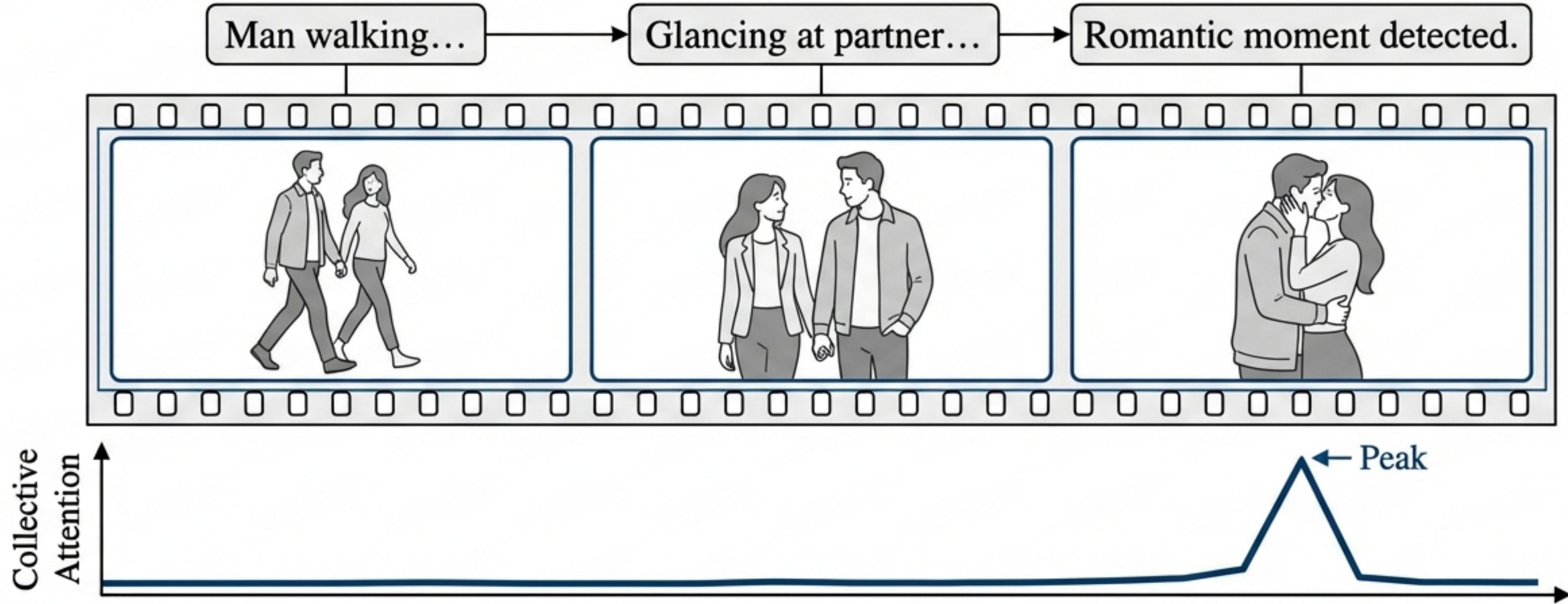
Modelling Long-Term Dependencies with Patch-Wise Transformers

- **Architecture:** Patch-Wise Segmentation applied to Transformers.
- **Function:** Normalization layers stabilize distribution shifts.
- **Application:** Predicting network states and user behaviors before they occur.



Video MLLMs & Spatiotemporal Reasoning.

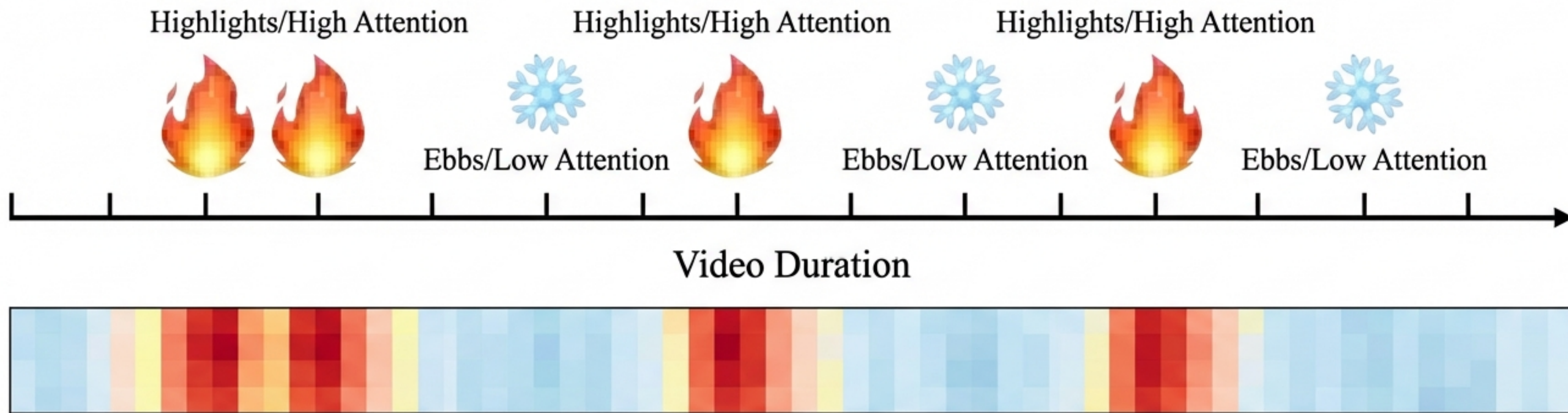
Capturing High Frame Density and Inter-Frame Dynamics.



- **Principle I:** High Frame Information Density.
- **Principle II:** High Inter-Frame Dynamics.
- **Innovation:** Reasoning about long-range context and physical laws in video.

Spatio-Temporal Attention & Collective Focus

Innovation: VideoLSTM & Attention Flow



Tracing collective attention surges in large-scale datasets (10,000+ videos)
to prioritize computational resources.

Logic-Constrained Visual Anomaly Detection

**Innovation: Symbolic Logic
+ Deep Learning**



Visual Input

$\text{In}(\text{Fire}, \text{GasStation}) = \text{Explosion}$

Inference:
EXPLOSION
(Anomaly)



Visual Input

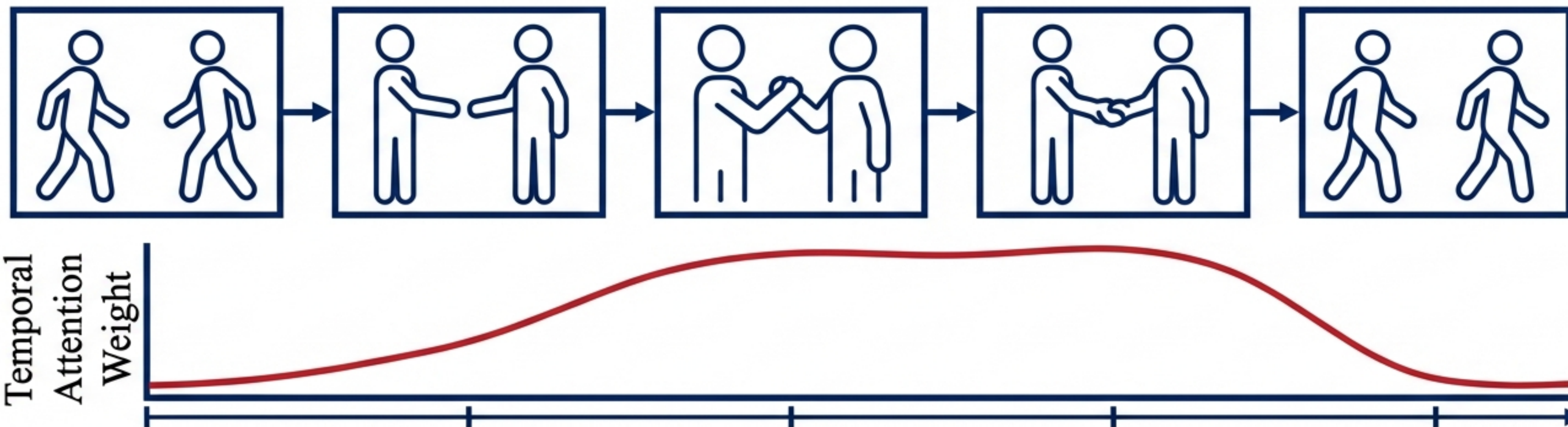
$\text{In}(\text{Fire}, \text{GasStation}) = \text{False}$

Inference:
FIRE
(Standard Event)

Integrating **symbolic logic** constraints (**Existence, Spatial**)
to reduce **false positives** in anomaly detection.

Dynamic Scene Understanding & Action Recognition

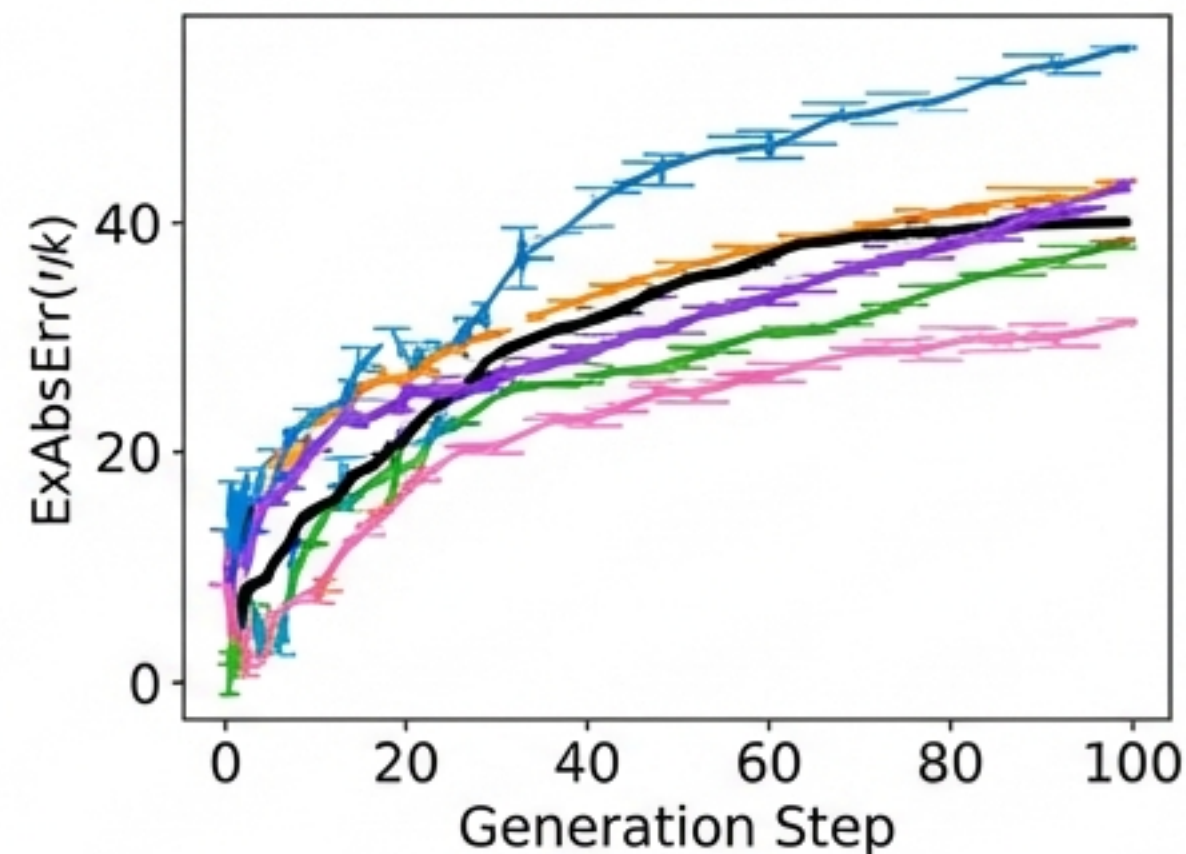
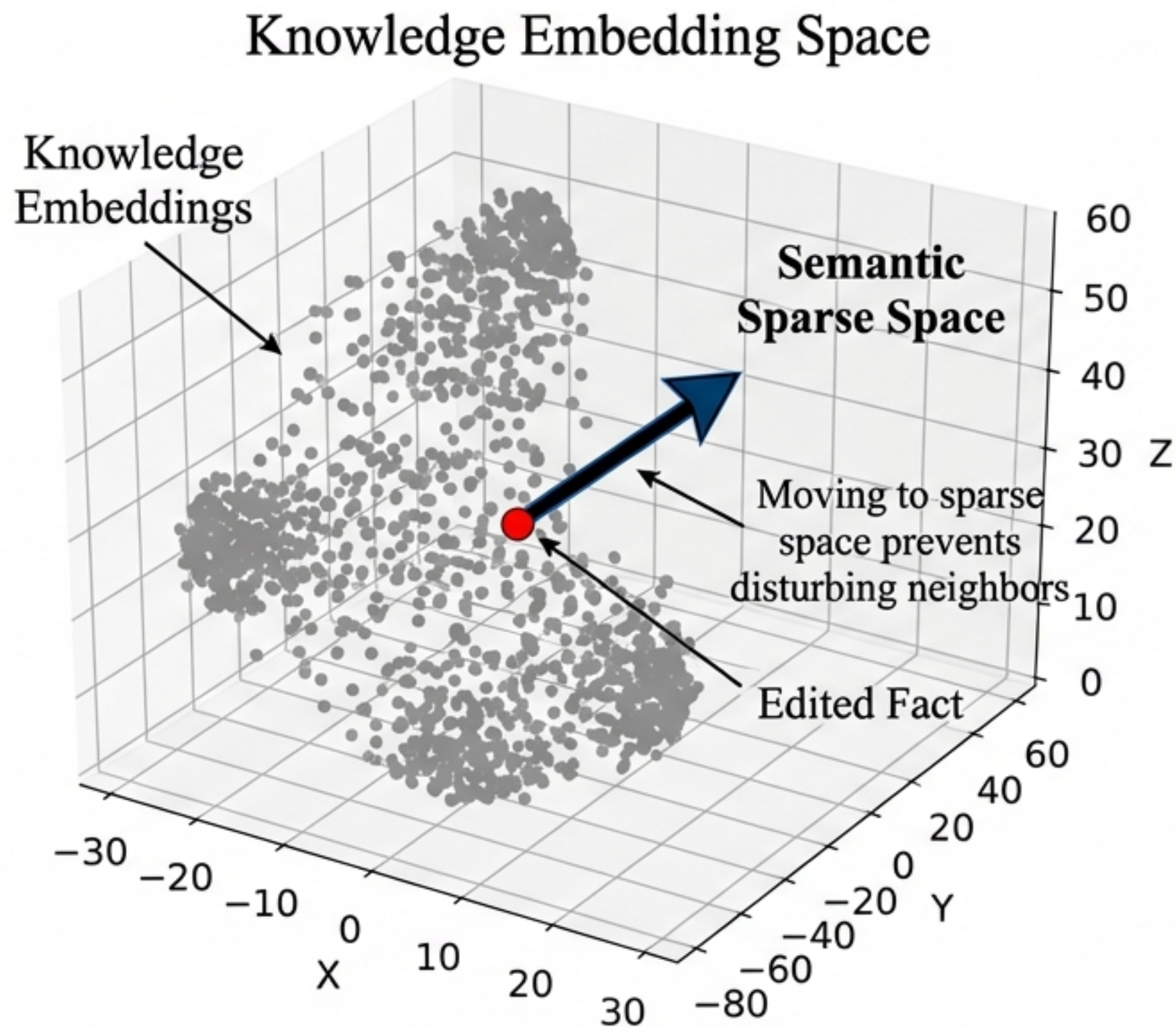
Innovation: Long-Short Term
Temporal Attention (LSCN)



Deep Bilinear Learning for RGB-D action recognition.
Identifying 'Richness' and 'Relevance' in video captioning.

Model Editing via Semantic Sparse Space (SSS).

Modifying Factual Knowledge Without Collateral Damage



- **Problem:** Editing LLMs often damages unrelated knowledge.
- **Solution:** The SSS Metric.
- **Mechanism:** Directs edited examples toward sparse semantic areas, preserving model integrity.

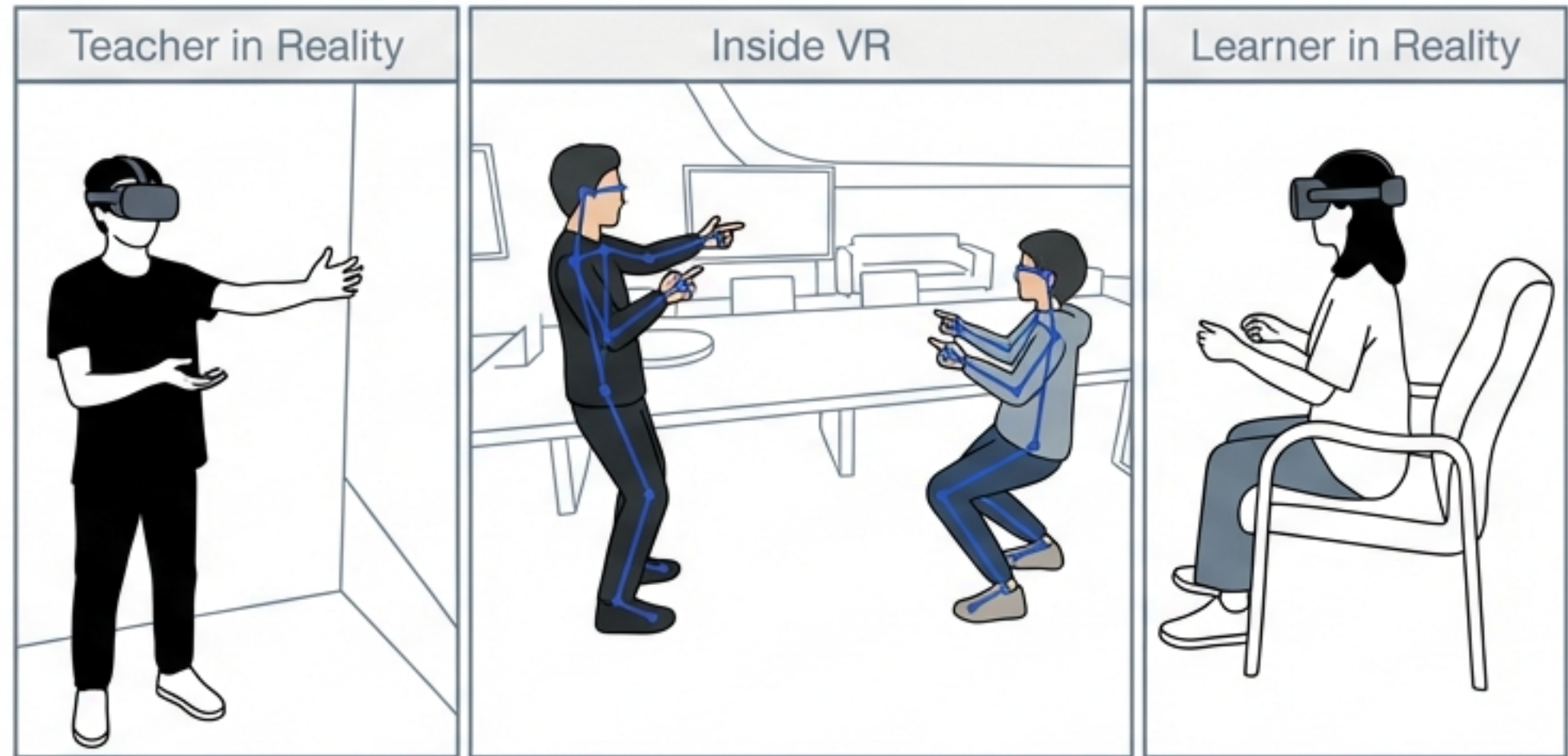
Embodied Intent: Cross-Reality Interaction

Core Capability: 3D Hand Pose Estimation.

Key Research: *Two Heads are Better than One: Image-Point Cloud Network* (AAAI Distinguished Paper).

Technique: Fusing 2D images with 3D point clouds for millimeter-level accuracy.

Goal: Controller-free, natural intent interaction.



Embodied Intent: Naked-Hand XR Interaction

Innovation:
MegoTrack & FreeWriting

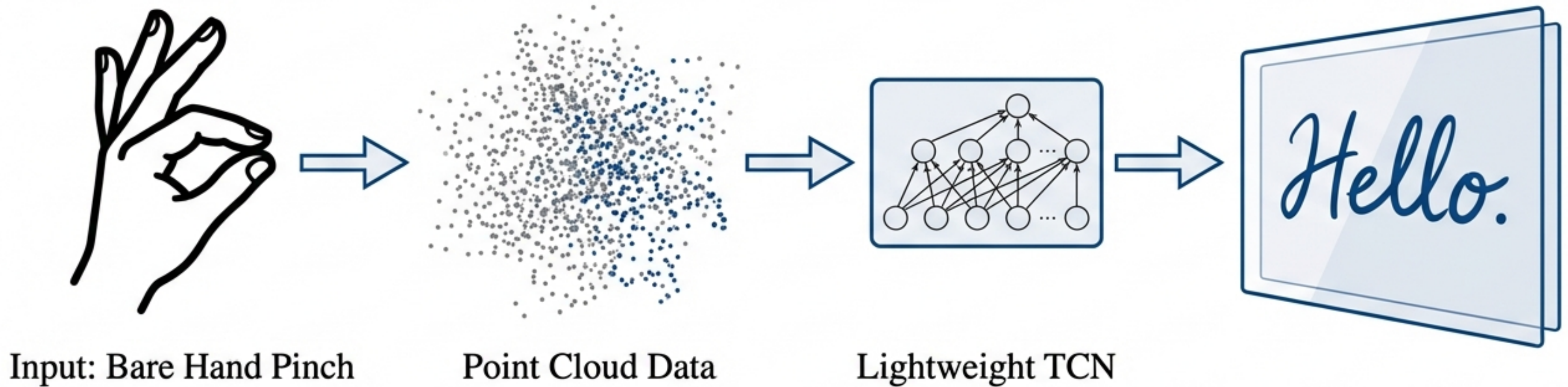


$$\begin{aligned} \sum_{i=1}^n (P_i \times W_i) + \int_0^x f(x) dx &= \text{Pose}(t) \\ + \sum_{k=0}^n (P_i \times i) &= \frac{\partial(xW_i)}{\partial t} \cdot j(r, xi) \\ + \int_0^x f(x) dx &= \text{Pose}(t) \end{aligned}$$

MegoTrack: Winner of ICCV Hands 2025 Challenge. Utilizing pre-calibrated hand shapes for high-precision dual-hand tracking without controllers.

InkFlow: Natural Mid-Air Handwriting in MR.

Controller-Free Interaction via Lightweight TCN Models



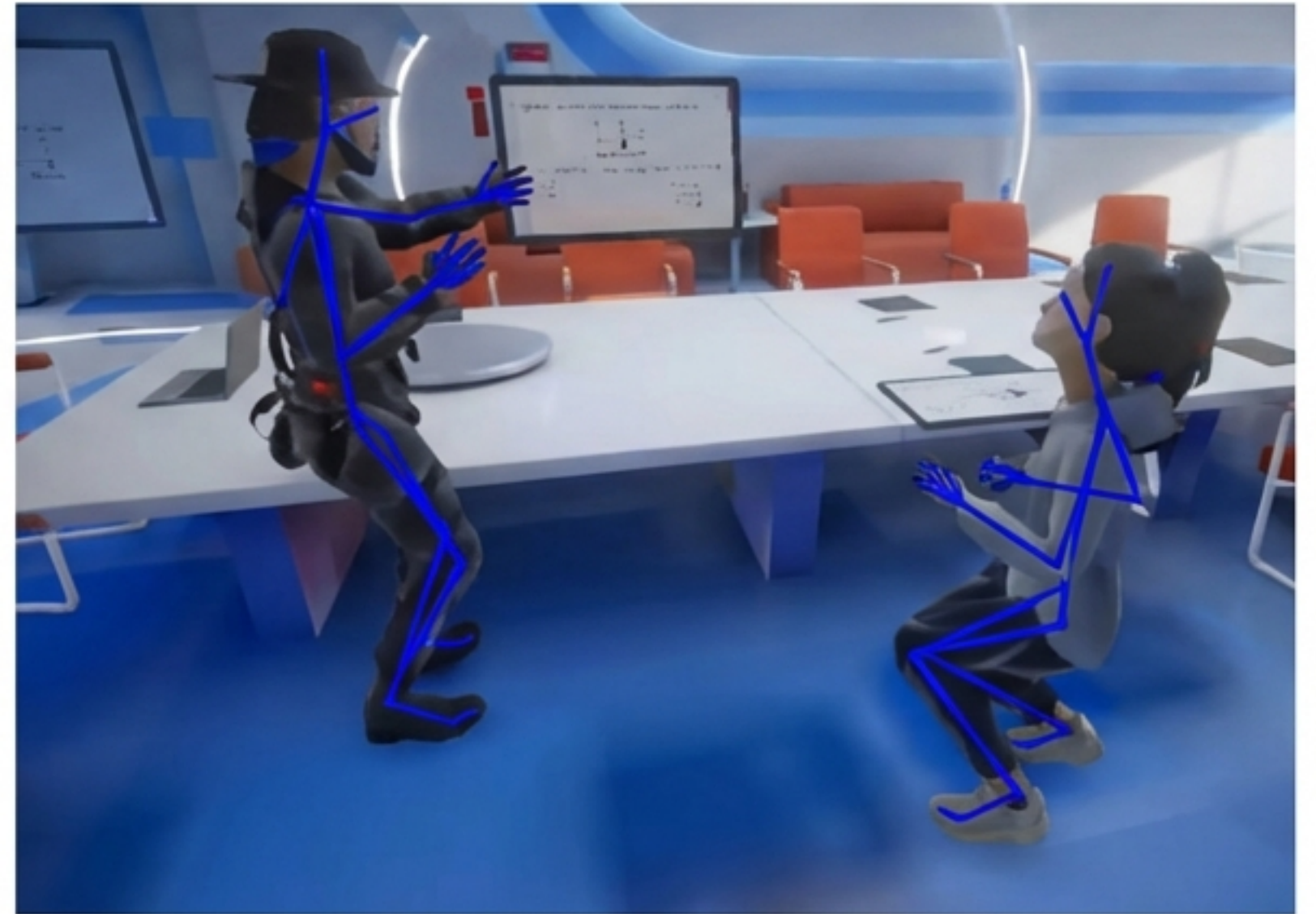
- **Innovation:** Eliminates controllers and proximal planes.
- **Gesture:** Uses 'Pinch-Release' for intuitive auto-annotation.
- **Architecture:** Depth-separable Temporal Convolutional Network (TCN) for headset efficiency.

Collaborative Synergy in the Metaverse

Study: Collaborative Communication Efficiency in Virtual Teaching.

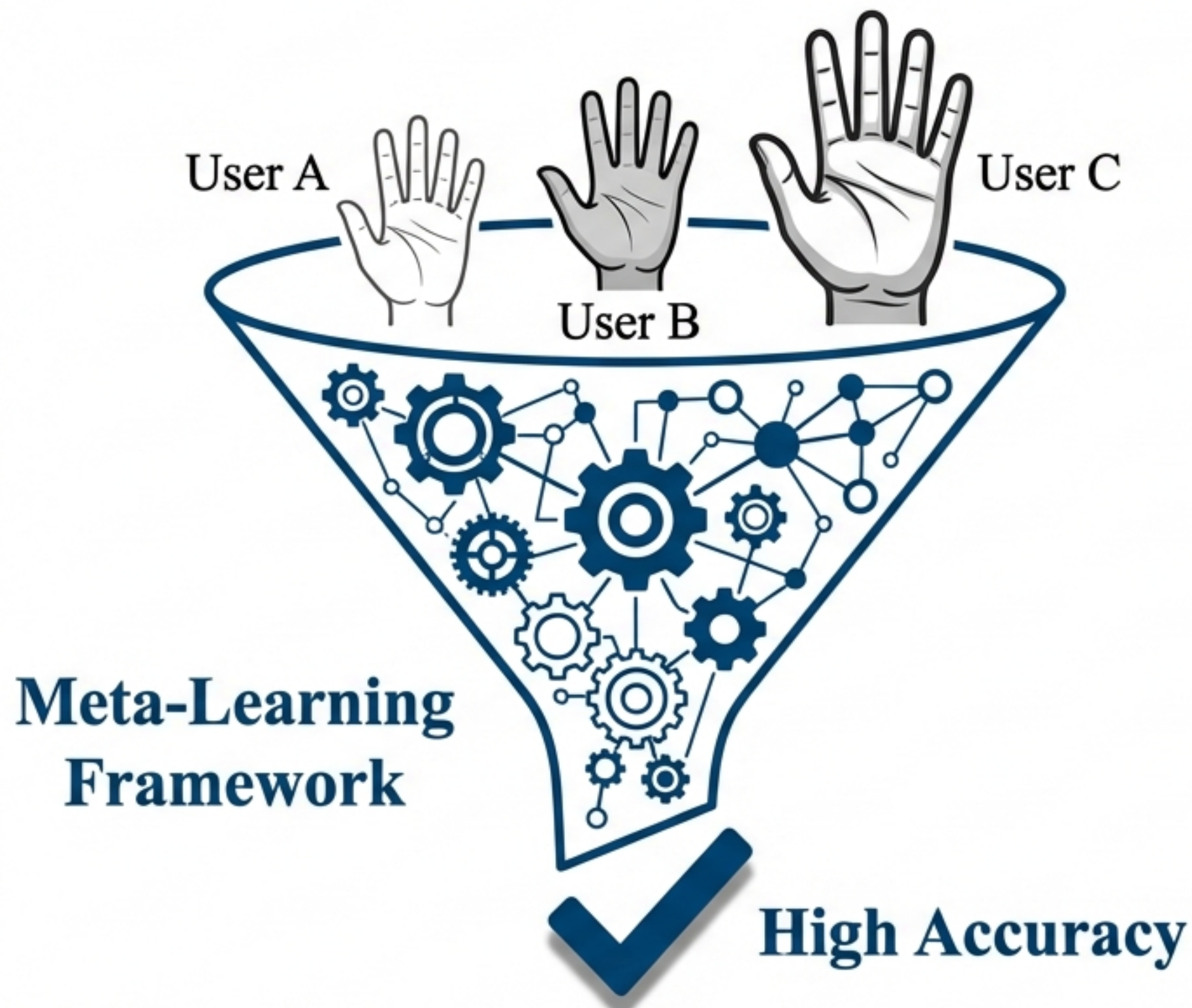
Findings: Natural 'Bare Hand' interaction significantly reduces cognitive load compared to controllers.

Result: Faster knowledge transfer in Physics, Chemistry, and Geography experiments.



Cross-User Generalization & Meta-Learning

Rapid Personalization for Diverse Hand Biometrics

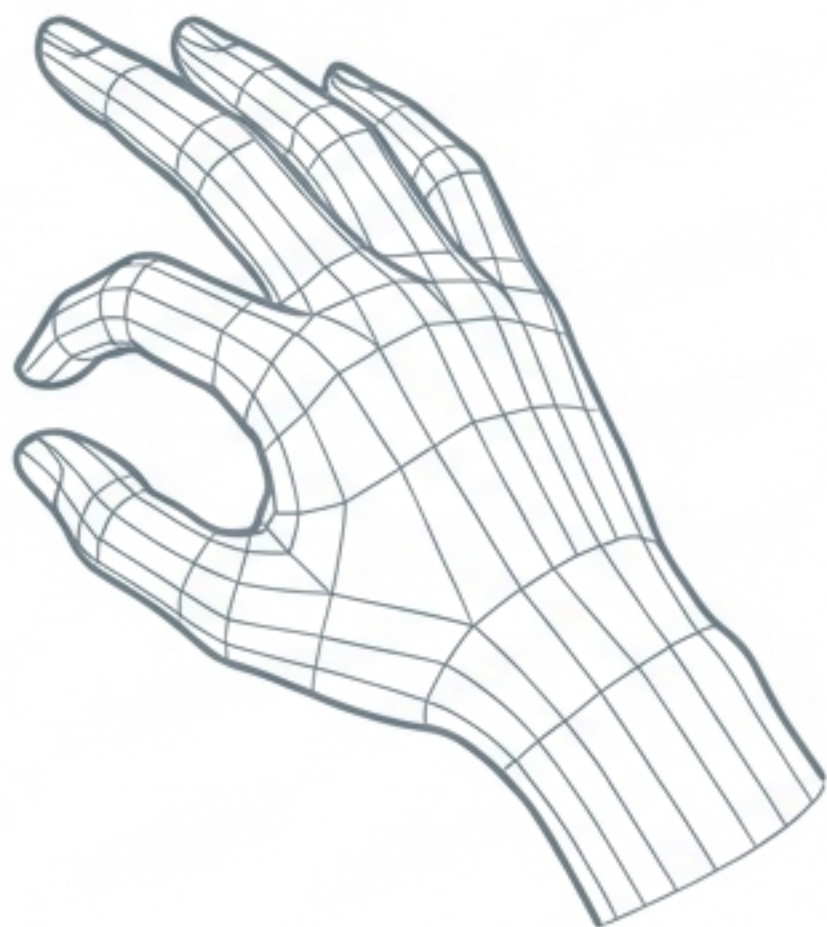


- **Challenge:** Unique hand shapes and writing styles per user.
- **Solution:** Meta-Learning with domain generalization.
- **Impact:** Rapid adaptation to new users with minimal data (Cold Start Solution).

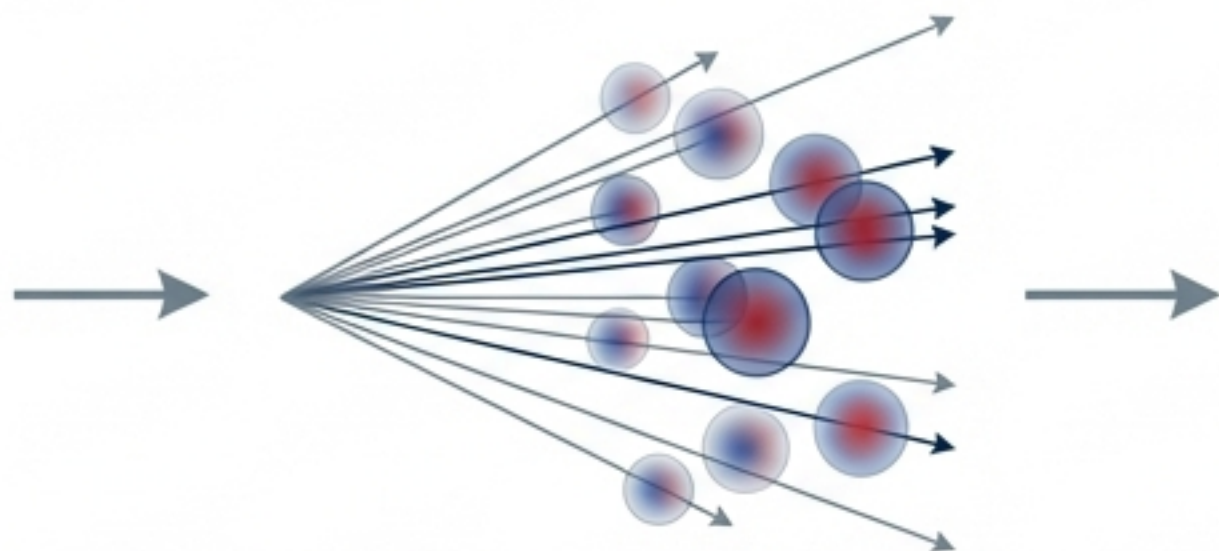
Generative Interaction & Controllable Diffusion

Innovation: Content Co-Awareness Latent Diffusion

Controllable Generation



User Gesture (Input)



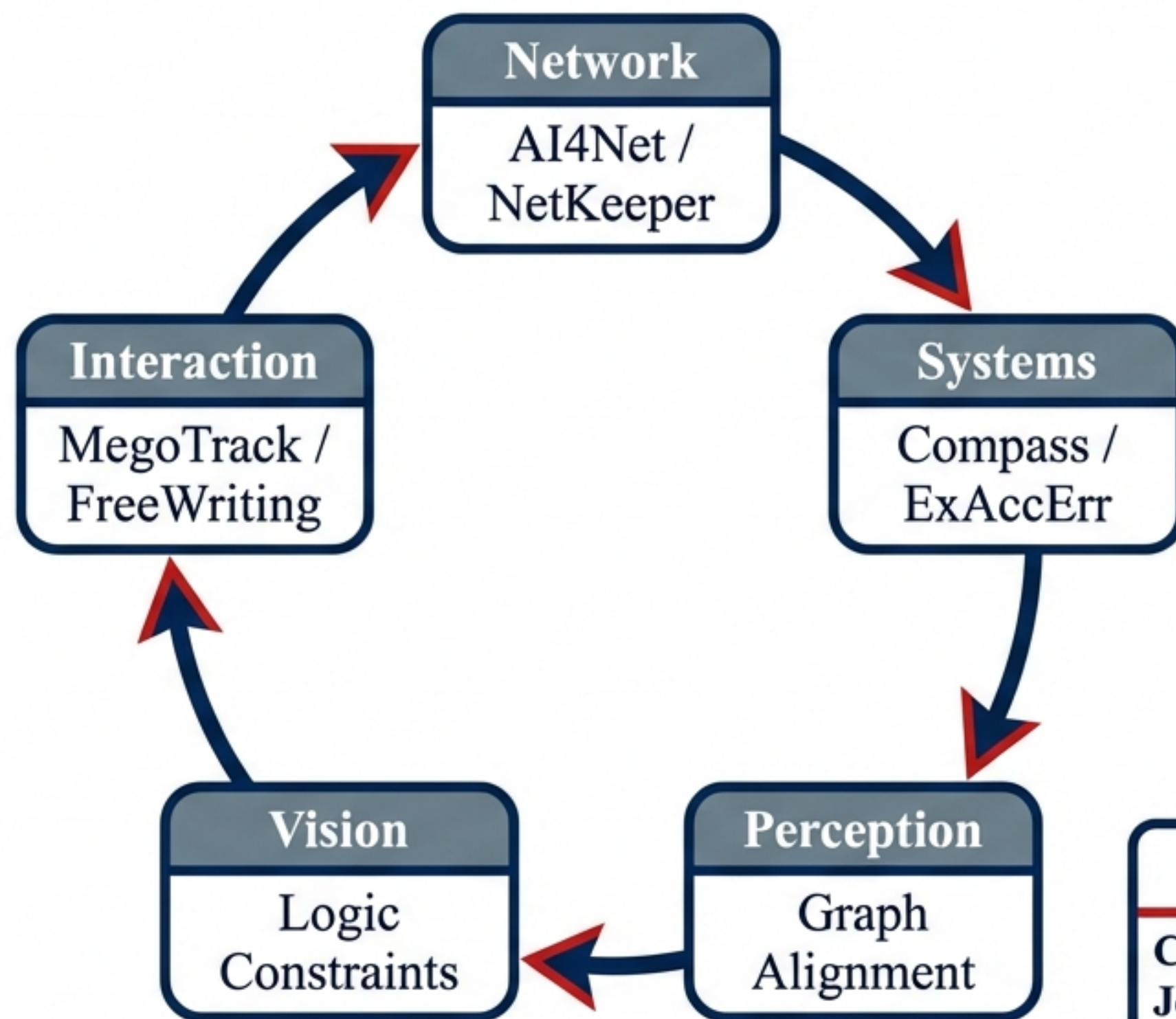
Gaussian Opacity Fields



Diffusion Output

Generating digital responses that respect physical constraints. CVPR 2026.

The iLink Ecosystem: A Closed-Loop Summary



Call to Action

Code: github.com/OpenNetAI
Join Us: NIRC - The Big Intelligence

Selected Premier Publications

Systems & Networking

USENIX NSDI

ACM SIGCOMM

USENIX ATC

IEEE/ACM ToN

AI & Vision

CVPR

ICCV

AAAI

IEEE TIP

HCI & Multimedia

ACM MM

CHI

CSCW

First BUPT papers in NSDI, ASPLOS, SIGMOD



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Building the Intelligent Link for the Future.